1.17 Proportion

Chapter 1 · Original slide 17 · 4: Negative and Fractional Indices

$a^0 = 1$
$a^{\frac{1}{m}} = \sqrt[m]{a}$
$a^{\frac{n}{m}} = \left(\sqrt[m]{a}\right)^n$
$a^{-m} = \frac{1}{a^m}$

Note: $\sqrt{9}$ only means the positive square root of 9, i.e. 3 not -3.

Otherwise, what would be the point of the $\pm$ in the quadratic formula before the $\sqrt{b^2 - 4ac}$?

Prove that $x^{\frac{1}{2}} = \sqrt{x}$

Evaluate $27^{-\frac{1}{3}}$

Evaluate $32^{\frac{2}{5}}$

Simplify $\left(\frac{1}{9}x^6y\right)^{\frac{1}{2}}$

Evaluate $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$

If $b = \frac{1}{9}a^2$, determine $3b^{-2}$ in the form kb^n where k, n are constants.

Chapter 1 · Original slide 18 · Writing a surd using indices

If $9\sqrt{3} = 3^k$, determine the value of k.

The key here is to write everything as powers with a consistent base, in this case, 3.

Note: In algebra we like to avoid mixed numbers. So don't write $3^{2\frac{1}{2}}$

1.17 Proportion

Chapter 1 · Original slide 19 · Further Examples

[Edexcel IGCSE May14-4H] Given that $\left(2^{\frac{1}{2}}\right)^n = \frac{2^x}{8^y}$

Express n in terms of x and y.

Write 8 as a power of 2 (putting a bracket around it), for consistency of base with the other powers.

[Edexcel IAL C12 Jan 2019 Q2c] Given that $y = 2^x$, express the following in terms of y.

$\frac{1}{4^{2x-3}}$

Write your expression in its simplest form.

Chapter 1 · Original slide 20 · Test Your Understanding

Given $2^x \times 4^y = \frac{1}{2\sqrt{2}}$ express y as a function of x. (3)

1.17 Proportion

Chapter 1 · Original slide 21 · Exercise 1D

4 a Find the value of $81^{\frac{1}{4}}$. **b** Simplify $x(2x^{-\frac{1}{3}})^4$.

5 Given that $y = \frac{1}{8}x^3$ express each of the following in the form kx^n, where k and n are constants.
a $y^{\frac{1}{3}}$ **b** $\frac{1}{2}y^{-2}$

Challenge

Chapter 4 · Original slide 21 · GCSE RECAP: Reciprocal Graphs



Sketch $y = \frac{1}{x}$

Sketch $y = -\frac{3}{x}$

Note: The scaling caused by the 3 isn't observable for this graph in isolation because the axes have no scale. This will only be observation for multiple graphs on the same axes.

Notice the distance between this line and the x-axis (i.e. the line $y = 0$) gradually decreases as the lines go off towards infinity. The line $y = 0$ is known as an asymptote of the graph.

! An asymptote is a line which the graph approaches but never reaches.

Asymptotes of $y = \frac{a}{x}$:

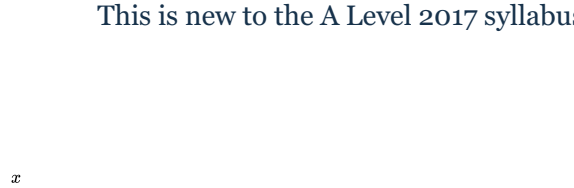
1.17 Proportion

Chapter 4 · Original slide 22 · Reciprocal Graphs



Sketch $y = \frac{3}{x^2}$

Hint: Note that anything squared will always be at least 0.



This is new to the A Level 2017 syllabus.

Sketch $y = -\frac{4}{x^2}$

Chapter 4 · Original slide 23 · Reciprocal Graphs

On the same axes, sketch $y = \frac{1}{x}$ and $y = \frac{3}{x}$

$y = \frac{3}{x}$

$y = \frac{1}{x}$

y

The y value for $y = \frac{3}{x}$ will be 3 times greater than $y = \frac{1}{x}$

x

1.17 Proportion

Chapter 4 · Original slide 24 · Exercise 4C – Pure Y1 page 67

1 Use a separate diagram to sketch each pair of graphs.

a $y = \frac{2}{x}$ and $y = \frac{4}{x}$ **b** $y = \frac{2}{x}$ and $y = -\frac{2}{x}$ **c** $y = -\frac{4}{x}$ and $y = -\frac{2}{x}$
d $y = \frac{3}{x}$ and $y = \frac{8}{x}$ **e** $y = -\frac{3}{x}$ and $y = -\frac{8}{x}$

Chapter 4 · Original slide 25 · Exercise 4C – Pure Y1 page 67

2 Use a separate diagram to sketch each pair of graphs.
a $y = \frac{2}{x^2}$ and $y = \frac{5}{x^2}$ **b** $y = \frac{3}{x^2}$ and $y = -\frac{3}{x^2}$ **c** $y = -\frac{2}{x^2}$ and $y = -\frac{6}{x^2}$